

Gramatyka

Rozważana gramatyka (z rozszerzonym symbolem startowym E'):

1. $E' \rightarrow E$
2. $E \rightarrow E \text{ or } T$
3. $E \rightarrow T$
4. $T \rightarrow T \text{ and } F$
5. $T \rightarrow F$
6. $F \rightarrow \text{not } F$
7. $F \rightarrow (E)$
8. $F \rightarrow \text{true}$
9. $F \rightarrow \text{false}$

Zbiory sytuacji i przejścia goto:

I_0 :

$$\begin{array}{c}
 E \\
 E \\
 T \\
 T \\
 F \rightarrow \cdot \text{not } F \\
 F \rightarrow \cdot (E) \\
 F \rightarrow \cdot \text{true} \\
 F \rightarrow \cdot \text{false}
 \end{array}
 \left. \begin{array}{l}
 \\
 \\
 \rightarrow \cdot T \text{ and } F \\
 \rightarrow \cdot F \\
 \\
 \\
 \end{array} \right\} \mu_F
 \left. \begin{array}{l}
 \\
 \\
 \\
 \\
 \\
 \\
 \end{array} \right\} \mu_T
 \left. \begin{array}{l}
 E' \rightarrow \cdot E \\
 \rightarrow \cdot E \text{ or } T \\
 \rightarrow \cdot T \\
 \\
 \\
 \\
 \end{array} \right\} \mu_E$$

$$\begin{aligned}
 &goto(I_0, E) = I_1 \\
 &E' \rightarrow E \cdot \\
 &E \rightarrow E \cdot \text{ or } T
 \end{aligned}$$

$$\begin{aligned}
 &goto(I_0, T) = I_2 \\
 &E \rightarrow T \cdot \\
 &T \rightarrow T \cdot \text{ and } F
 \end{aligned}$$

$$\begin{aligned}
 &goto(I_0, F) = I_3 \\
 &T \rightarrow F \cdot
 \end{aligned}$$

$goto(I_0, not) = I_4$
 $F \rightarrow not \cdot F$
 $F \rightarrow \cdot not F$
 $F \rightarrow \cdot (E)$
 $F \rightarrow \cdot true$
 $F \rightarrow \cdot false$

$goto(I_0, ()) = I_5$
 $F \rightarrow (\cdot E)$
 μ_E

$goto(I_0, true) = I_6$
 $F \rightarrow true \cdot$

$goto(I_0, false) = I_7$
 $F \rightarrow false \cdot$

$goto(I_1, or) = I_8$
 $E \rightarrow E or \cdot T$
 μ_T

$goto(I_2, and) = I_9$
 $T \rightarrow T and \cdot F$

$goto(I_4, F) = I_{10}$
 $F \rightarrow not F \cdot$

$goto(I_4, not) : I_4$
 $goto(I_4, ()) : I_5$
 $goto(I_4, true) : I_6$
 $goto(I_4, false) : I_7$
 $goto(I_5, E) = I_{11}$
 $F \rightarrow (E \cdot)$
 $E \rightarrow E \cdot or T$

$goto(I_5, T) : I_2$
 $goto(I_5, F) : I_3$
 $goto(I_5, not) : I_4$
 $goto(I_5, ()) : I_5$
 $goto(I_5, true) : I_6$
 $goto(I_5, false) : I_7$
 $goto(I_8, T) = I_{12}$
 $E \rightarrow or T \cdot$
 $T \rightarrow T \cdot and F$

$goto(I_8, F) : I_3$
 $goto(I_8, not) : I_4$
 $goto(I_8, () : I_5$
 $goto(I_8, true) : I_6$
 $goto(I_8, false) : I_7$
 $goto(I_9, F) = I_{13}$
 $T \rightarrow T \text{ and } F.$

$goto(I_9, not) : I_4$
 $goto(I_9, () : I_5$
 $goto(I_9, true) : I_6$
 $goto(I_9, false) : I_7$
 $goto(I_{11},)) = I_{14}$
 $F \rightarrow (E).$

$goto(I_{11}, or) : I_8$
 $goto(I_{12}, and) : I_9$

Zbiory FIRST

$$\begin{aligned}
 FIRST(F) &= \{not, (, true, false\} \\
 FIRST(T) &= FIRST(F) \\
 FIRST(E) &= FIRST(T)
 \end{aligned}$$

Zbiory FOLLOW

$$\begin{aligned}
 FOLLOW(E) &= \{\$,), or\} \\
 FOLLOW(T) &= \{\$,), or, and\} \\
 FOLLOW(F) &= \{\$,), or, and\}
 \end{aligned}$$

Tablica SLR

Terminale: or, and, not, (,), true, false, \$
 Nieterminale: E, T, F

Stan	not	(	true	false	)	or	and	\$	E	T	F
0	s4	s5	s6	s7					1	2	3
1						s8		acc			
2					r3	r3	s9	r3			
3					r5	r5	r5	r5			
4	s4	s5	s6	s7					11		10
5	s4	s5	s6	s7						2	3
6					r8	r8	r8	r8			
7					r9	r9	r9	r9			
8	s4	s5	s6	s7						2	3
9	s4	s5	s6	s7							13
10					r6	r6	r6	r6			
11					s14	s8					
12					r2	r2	s9	r2			
13					r4	r4	r4	r4			
14					r7	r7	r7	r7			